CS-515 Advanced CS Practicum Lab

Objective 2

t26049 - Yashank

Problem Statement

In computational graph theory, determining exact two-color Ramsey numbers $R(s,t)$ is notoriously difficult due to combinatorial explosion. The search space for evaluating all possible red/blue edge colorings of a complete graph K_N grows exponentially at a rate of $2^{N(N-1)/2}$, making naive brute-force search computationally intractable even for small values of N. The objective of this project is to design and implement a highly optimized algorithmic solver that systematically finds counterexamples to establish lower bounds, or strictly proves exact Ramsey numbers. This requires overcoming the exponential time-complexity barrier by implementing advanced algorithmic techniques, including bitwise clique-detection, strict degree-bound pruning, symmetry breaking, and fast heuristic pattern-matching via circulant graphs.

Algorithm

```
ALGORITHM Find_Ramsey_Number(s, t, budget)

U = Calculate_Upper_Bound(s, t)

// Pre-compute largest pattern-based counterexample (Threshold: N <= 31)

big_pattern = Find largest Circulant_Search for N from 31 down to 5

FOR N = 1 TO U:

counterexample = NULL

// 1. Fast Heuristic Search (Threshold: 5 <= N <= 31)

IF 5 <= N <= 31 THEN
```

```
counterexample = Circulant_Search(N, s, t)

// 2. Exhaustive Backtracking (If heuristic fails)

IF counterexample IS NULL THEN

counterexample = Exhaustive_Backtracking(N, s, t, budget)

// Note: Internally uses Bitwise Clique & Degree-Bound Pruning

// 3. Budget Fallback (If backtracking runs out of time)

IF counterexample IS NULL AND budget_is_exhausted THEN

IF N <= size(big_pattern) THEN

counterexample = First N vertices of big_pattern

// 4. Evaluate Checkpoint

IF counterexample IS NULL THEN

// No counterexample exists -> Clique is mathematically forced

RETURN "R(s, t) = " + N

RETURN "R(s, t) = " + U
```

Result

```
=== R(2,2): red K_2 vs blue K_2 ===
  N= 0  counterexample found (backtracking, 0 nodes, 0.0s)
  N= 1  counterexample found (backtracking, 0 nodes, 0.0s)
  N= 2  NO counterexample: exhaustive search of 1 nodes (0 clique prunes, 1 degree prunes, 0.0s)
  => every red/blue colouring of K_2 has a red K_2 or a blue K_2

R(2,2) = 2
Largest counterexample: K_1 with no red K_2 and no blue K_2:
     0
   0  .
```

- for R(2,2) number of minimum vertices require are 2 as we can say 2 vertices will either know each other or will not know each other

```
=== R(3,3): red K_3 vs blue K_3 ===
  N= 0  counterexample found (backtracking, 0 nodes, 0.0s)
  N= 1  counterexample found (backtracking, 0 nodes, 0.0s)
  N= 2  counterexample found (backtracking, 1 nodes, 0.0s)
  N= 3  counterexample found (backtracking, 4 nodes, 0.0s)
  N= 4  counterexample found (backtracking, 8 nodes, 0.0001s)
  N= 5  counterexample found (circulant, 0.0001s)
  N= 6  NO counterexample: exhaustive search of 113 nodes (3 clique prunes, 54 degree prunes, 0.0001s)
  => every red/blue colouring of K_6 has a red K_3 or a blue K_3

R(3,3) = 6
Largest counterexample: K_5 with no red K_3 and no blue K_3:
     0  1  2  3  4
  0  .  R  B  B  R
  1  R  .  R  B  B
  2  B  R  .  R  B
  3  B  B  R  .  R
  4  R  B  B  R  .
```

- for R(3,3) algorithm was able to find counter Example till N = 5 but for N = 6 even using exhaustive search we could not find the counterExample so we can safely say 6 is the required N for R(3,3)

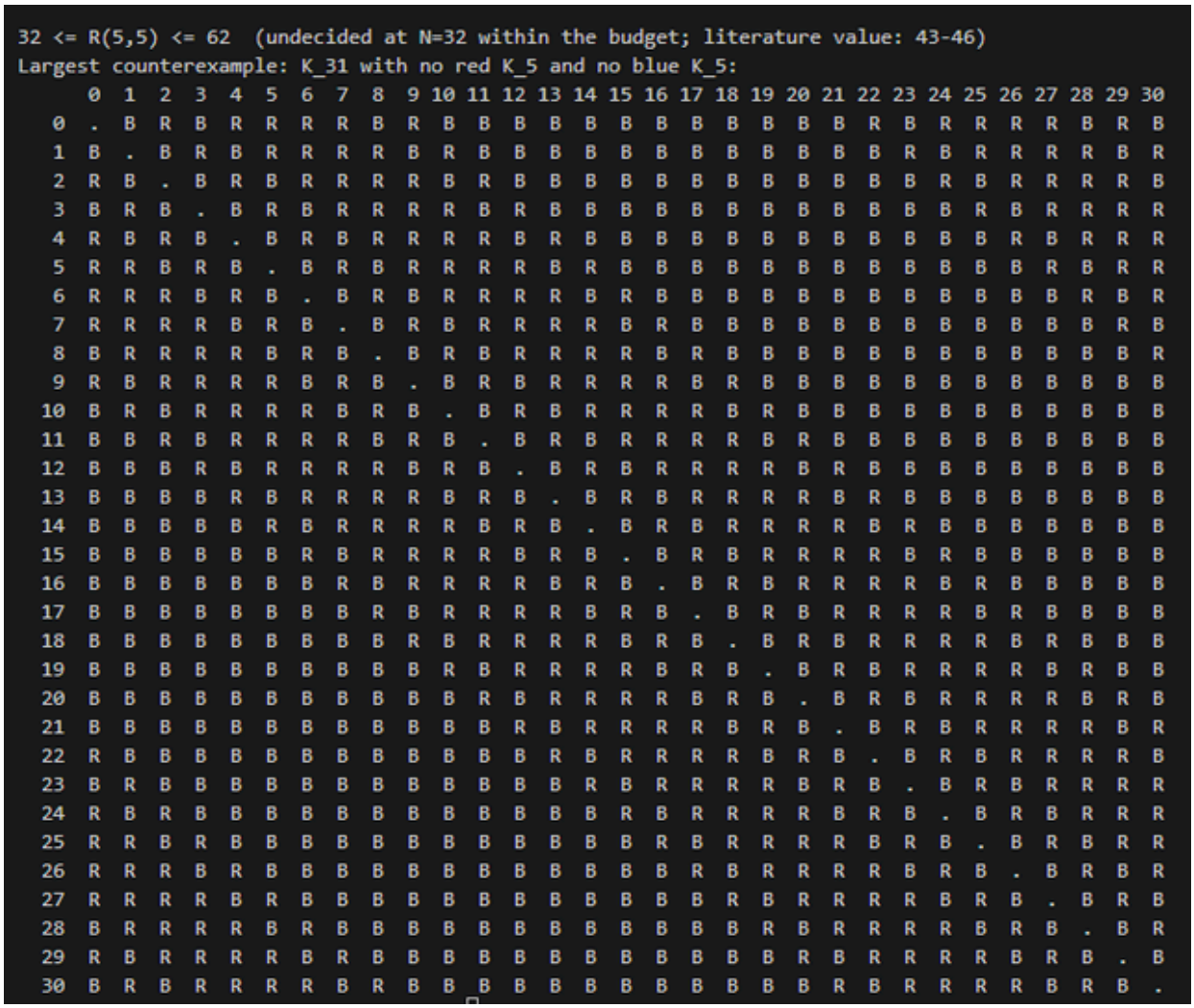
```
  N= 8  counterexample found (circulant, 0.0002s)
  N= 9  counterexample found (circulant, 0.0002s)
  N=10  counterexample found (circulant, 0.0002s)
  N=11  counterexample found (circulant, 0.0003s)
  N=12  counterexample found (circulant, 0.0009s)
  N=13  counterexample found (circulant, 0.001s)
  N=14  counterexample found (restriction of K_17, 5000001 nodes, 6.0737s)
  N=15  counterexample found (restriction of K_17, 5000001 nodes, 6.7248s)
  N=16  counterexample found (restriction of K_17, 5000001 nodes, 6.4501s)
  N=17  counterexample found (circulant, 0.0125s)
  N=18  NO counterexample: forced by the bound R(4,4) <= R(3,4) + R(4,3) = 18 (exhaustive search would exceed the budget)

R(4,4) = 18
Largest counterexample: K_17 with no red K_4 and no blue K_4:
     0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
  0  .  B  B  R  B  R  R  R  B  B  R  R  R  B  R  B  B
  1  B  .  B  B  R  B  R  R  R  B  B  R  R  R  B  R  B
  2  B  B  .  B  B  R  B  R  R  R  B  B  R  R  R  B  R
  3  R  B  B  .  B  B  R  B  R  R  R  B  B  R  R  R  B
  4  B  R  B  B  .  B  B  R  B  R  R  R  B  B  R  R  R
  5  R  B  R  B  B  .  B  B  R  B  R  R  R  B  B  R  R
  6  R  R  B  R  B  B  .  B  B  R  B  R  R  R  B  B  R
  7  R  R  R  B  R  B  B  .  B  B  R  B  R  R  R  B  B
  8  B  R  R  R  B  R  B  B  .  B  B  R  B  R  R  R  B
  9  B  B  R  R  R  B  R  B  B  .  B  B  R  B  R  R  R
 10  R  B  B  R  R  R  B  R  B  B  .  B  B  R  B  R  R
 11  R  R  B  B  R  R  R  B  R  B  B  .  B  B  R  B  R
 12  R  R  R  B  B  R  R  R  B  R  B  B  .  B  B  R  B
 13  B  R  R  R  B  B  R  R  R  B  R  B  B  .  B  B  R
 14  R  B  R  R  R  B  B  R  R  R  B  R  B  B  .  B  B
 15  B  R  B  R  R  R  B  B  R  R  R  B  R  B  B  .  B
 16  B  B  R  B  R  R  R  B  B  R  R  R  B  R  B  B  .
```

- similarly for R(4,4) as we can notice how fast we were able to find the result using circulant algorithm till N=13 though it took some time to find the result for n = 14 to 16 algorithms was collectively able to determine N = 18 using the mathematical formula for upper bound in ramsey search

- R(s,t) <= R(s-1,t) + R(s,t-1), as exhaustive search must not exceed this budget

```
N= 0  counterexample found (backtracking, 0 nodes, 0.0s)
N= 1  counterexample found (backtracking, 0 nodes, 0.0s)
N= 2  counterexample found (backtracking, 1 nodes, 0.0s)
N= 3  counterexample found (backtracking, 3 nodes, 0.0s)
N= 4  counterexample found (backtracking, 6 nodes, 0.0s)
N= 5  counterexample found (circulant, 0.0s)
N= 6  counterexample found (circulant, 0.0s)
N= 7  counterexample found (circulant, 0.0001s)
N= 8  counterexample found (circulant, 0.0001s)
N= 9  counterexample found (circulant, 0.0001s)
N=10  counterexample found (circulant, 0.0002s)
N=11  counterexample found (circulant, 0.0002s)
N=12  counterexample found (circulant, 0.0002s)
N=13  counterexample found (circulant, 0.0003s)
N=14  counterexample found (circulant, 0.0005s)
N=15  counterexample found (circulant, 0.0007s)
N=16  counterexample found (circulant, 0.0008s)
N=17  counterexample found (circulant, 0.0011s)
N=18  counterexample found (circulant, 0.0016s)
N=19  counterexample found (circulant, 0.0011s)
N=20  counterexample found (circulant, 0.004s)
N=21  counterexample found (circulant, 0.0053s)
N=22  counterexample found (circulant, 0.0088s)
N=23  counterexample found (circulant, 0.0118s)
N=24  counterexample found (circulant, 0.0119s)
N=25  counterexample found (circulant, 0.0145s)
N=26  counterexample found (circulant, 0.0165s)
N=27  counterexample found (circulant, 0.0327s)
N=28  counterexample found (circulant, 0.0584s)
N=29  counterexample found (circulant, 0.1099s)
N=30  counterexample found (circulant, 0.1157s)
N=31  counterexample found (circulant, 0.1401s)
N=32  search budget exhausted after 5000001 nodes - undecided
```



- R(5,5) is still an open problem we were able to find counter example till N = 31 because of circulant search but because we have an upper bound till n=31 on circulant search we have to use exhaustive backtracking which used up all the budget and was still unable to find counterExample for N=32 and because we exited because of budget we can safely say that for N=32 it is undecided

- there is a upper bound on circulant search because it requires 2^N//2 test operations to run and a computer can evaluate upto 32000 test operation within fraction of a second but after 2^15 things go beyond this limit taking a lot of extra time